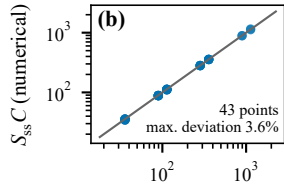
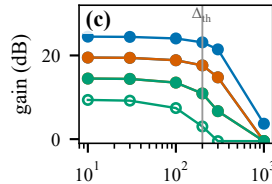


measurement time (ms)

$\bar{n} = 10^6$ $\bar{n} = 10^9$
 $\bar{n} = 10^7$ $\bar{n} = 10^{10}$
 $\bar{n} = 10^8$ twisting, Lorentzian to Voigt

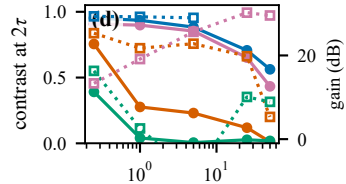


$4g\sqrt{\eta_d \bar{n}}/\kappa$



$\Delta/2\pi$ (MHz)

$\Delta/2\pi = 30$ MHz
 $\Delta/2\pi = 200$ MHz
 $\Delta/2\pi = 300$ MHz
 $\Delta/2\pi = 1000$ MHz



echo half-time τ (ms)

$\Delta/2\pi = 30$ MHz
 $\Delta/2\pi = 200$ MHz
 $\Delta/2\pi = 300$ MHz
 $\Delta/2\pi = 1000$ MHz